Mini-project — Compressed Air Energy Storage (CAES)

System design and simulation

*Revised specification: additions and corrections identified by fully modelling the six variants are shown **in red**.*

1. Introduction

Compressed Air Energy Storage (CAES) stores electricity as the potential energy of pressurised air. During *charge*, off-peak or surplus electricity drives a compressor train that pressurises air into a reservoir (an underground cavern or a pressure vessel). During *discharge*, the stored air is released, heated, and expanded through a turbine that drives a generator, returning electricity to the grid at peak hours.

CAES is one of the few technologies suitable for **large-scale, long-duration** electricity storage, alongside pumped hydro. It is used for peak shaving, grid balancing, and the integration of variable renewable generation (wind, solar). Its central engineering challenge is the **heat of compression**: compressing air heats it strongly, and expanding it cools it. How this heat is managed (rejected, stored, or supplied by fuel) largely determines the round-trip performance, and is exactly what distinguishes the variants studied here.

2. The six variants

The project explores **three architectures**, each with **two reservoir types**, giving six systems. **Each group studies one variant.** All variants share the same specification (Section 3) so that, at the end, the groups can **compare their results** on a common basis.

Architecture	**Constant-volume reservoir**	**Constant-pressure reservoir**
Single stage, no thermal storage	Variant 1a	Variant 1b
Single stage, with thermal storage	Variant 2a	Variant 2b
Two-stage (intercooling + reheat)	Variant 3a	Variant 3b

The **architecture** sets how the compression heat is handled: rejected to ambient (1), stored and reused (2, thermal storage), or reduced by staging with intercooling and recovered partly through reheat (3). The **reservoir type** sets the storage behaviour: constant volume (the pressure slides as the store fills and empties) or constant pressure (the volume varies, e.g. with hydraulic compensation).

For every variant the work is the same two-step request, as in the group's other projects:

1. **Design mode** — size the components to meet the common specification (in particular the **heat-exchanger areas** and the **storage volume**), at the common sizing point $P = P_{\max} = 16$ bar;
2. **Simulation mode** — with the sized system (the heat-exchanger areas fixed from the design step), simulate a full charge/discharge cycle and evaluate its performance.

3. Common specification and comparison criteria

All values below are **identical for the six variants**. Fixing them (especially the isentropic efficiencies) is what makes the final comparison meaningful.

Reference conditions and target

Quantity	Value	Note
Working fluid	Air	real-gas properties (CoolProp "Air")
Ambient temperature T_0	15 °C	air intake and heat-rejection reference
Ambient pressure P_0	1.013 bar	compressor inlet / turbine outlet
Discharge power	1 MW	at the turbine/generator shaft; held **constant** (the discharge flow is regulated to it)
Discharge duration	4 h	⇒ 4 MWh delivered per cycle
Charge duration	4 h	sets the charge mass flow (= discharged mass / 4 h)
Maximum storage pressure $P_{\max}$	16 bar	stored air at T_0; end of discharge reference
Min. storage pressure $P_{\min}$	8 bar	end of discharge, constant-volume variants (a)
Max. cold-air reject temp. $T_{\mathrm{amb,max}}$	50 °C	coolant air not rejected hotter than this (sets its flow)

Sizing point. All components are sized at $P = P_{\max} = 16$ bar, common to the six variants (for the constant-pressure variants the pressure is 16 bar throughout anyway).

Fixed component parameters

Parameter	Value	Note
Compressor isentropic efficiency $\eta_{is,c}$	0.85	same for every stage
Turbine isentropic efficiency $\eta_{is,t}$	0.88	same for every stage
Mechanical / generator efficiency	0.98	
Cooler pinch (inter/after-coolers)	10 K	air cooled towards $T_0 + 10$ K
Thermal-storage return pinch	10 K	variants 2a/2b
Two-stage pressure-ratio split	equal	$r_{\mathrm{stage}} = \sqrt{r_{\mathrm{overall}}}$ (variants 3a/3b)
Turbine inlet temperature (TIT)	550 °C	reached by combustion before each turbine
Fuel-to-electricity reference efficiency η_{f}	0.40	used to value fuel (see metric)
Film coefficient, air (gas) side	50 W/m^{2}K	used to turn a duty into an area
Film coefficient, water side	1000 W/m^{2}K	thermal-storage exchangers
Area ratio, air/air exchanger	1 : 1	equal areas
Area ratio, air/water exchanger	air 15 : water 1	extended (finned) air side

Thermal storage (variants 2 only). Two-tank sensible store using **pressurised liquid water** (≥ 60 bar, to stay liquid): cold tank at **25 °C**, hot tank at **250 °C**. The water mass (= water flow × 4 h) is reported as the **thermal-storage size**, an additional comparison criterion. See the modelling note in §3.3.

Modelling assumptions

- **Reservoir**: 0D (uniform pressure and temperature). The store is treated as **adiabatic during each phase** (its temperature is free, falling on discharge, rising on charge); it returns to T_0 only **between phases**, while at rest, through slow thermal losses. This return to T_0 at the hinge points is what closes the cycle.

- **Constant-volume reservoir**: the pressure slides between $P_{\max}$ and a minimum $P_{\min}$; the turbine sees this **sliding inlet pressure** (no throttling, to avoid destroying exergy). Take $P_{\min} = 8$ bar as the end-of-discharge pressure. These 16/8 bar bounds apply to the **discharge**. On charge the store starts at $p_1 \neq 8$ bar (the min-mass state warmed back to T_0) and the compressor must deliver up to $p_{\text{hot}} > 16$ bar (the hot end-of-charge pressure, ≈ 19 bar for 1a); the stored pressure only returns to 16 bar once cooled to T_0. Size the compressor to p_{hot}, not 16 bar.
- **Constant-pressure reservoir**: the pressure is held at $P_{\max}$ (variable volume); the turbine sees a **constant inlet pressure**.
- **Combustion** is modelled as a simple heat addition (bringing the air to the TIT before each turbine, no combustion chemistry): the duty is $Q = \dot{m}\,(h_{\text{TIT}} - h_{\text{in}})$ from real-gas enthalpy, **without** a constant-c_p assumption.
- **Thermal storage (variants 2) — only partial recovery.** The compression heat (≈ 1100 kW) exceeds what the discharge can reuse: the air can be pre-heated only to (water − 10 K pinch) ≈ 240 °C, i.e. ≈ 580 kW. **Size the water store by the discharge** (its reusable duty); on charge the air/water exchanger recovers only that amount and a **trim cooler rejects the rest** to bring the air to $T_0 + 10$ K before the air store. This keeps the two-tank water loop closed (same flow, 25 °C ↔ 250 °C).
- **Large-span coolers.** A single exchanger spanning $\approx 410 \to 25$ °C is split into **two exchangers in series** of half the area each (the single-zone method is inaccurate over such a span).
- Even without thermal storage, the compressed air is cooled before storage; the difference is only whether that heat is **recovered** (variants 2) or **rejected** (variants 1 and 3).

Comparison criteria

The common performance metric is a round-trip efficiency in which the fuel is valued at its electrical equivalent:

$$\eta = \frac{W_{\text{turbine}}}{W_{\text{compressor}} + \eta_{\text{f}}\, Q_{\text{combustion}}}, \qquad \eta_{\text{f}} = 0.40$$

This form is always finite (the denominator never vanishes) and puts electricity and fuel on the same footing. A value $\eta > 0.40$ means the system converts the fuel better than a 0.40-efficient reference plant would.

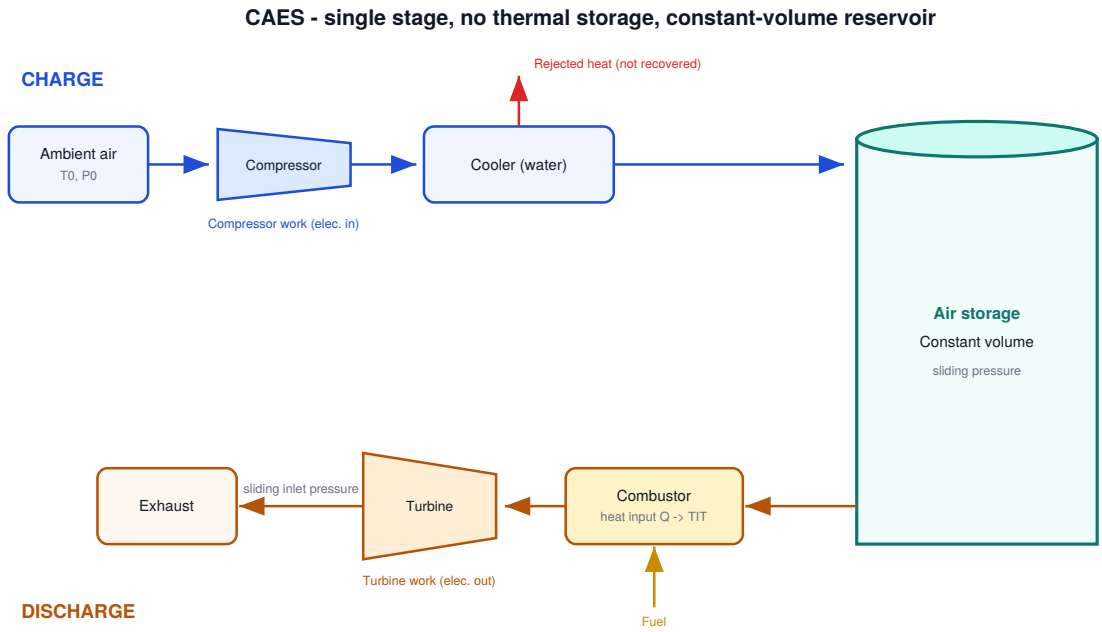
In addition to η, the groups compare:

- the **required storage volume** for the 4 MWh target, defined as the physical volume the vessel must hold: for a constant-volume store, the rigid vessel (cushion gas included); for a constant-pressure store, the maximum variable volume, evaluated at the warmest state (end of charge);
- the **heat-exchanger sizes** (areas) from the design step, compared as the **air-side area** A_{air} (the dimensioning surface), not the total of both sides;
- the **thermal-storage size** (water mass/volume), variants 2;
- the **maximum temperatures** reached (compressor outlet, TIT);
- the split between **electrical input** ($W_{\text{compressor}}$) and **fuel input** ($Q_{\text{combustion}}$).

4. Variant specifications

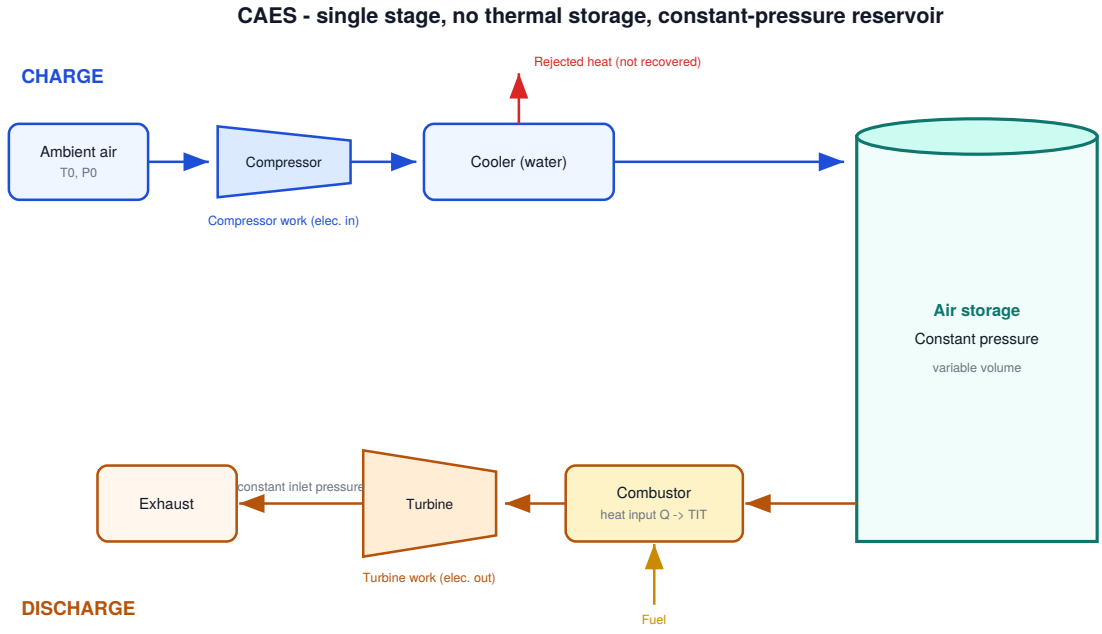
Each subsection shows the system a group must model. The common specification (Section 3) applies throughout; only the points below change.

Variant 1a — single stage, no thermal storage, constant-volume reservoir



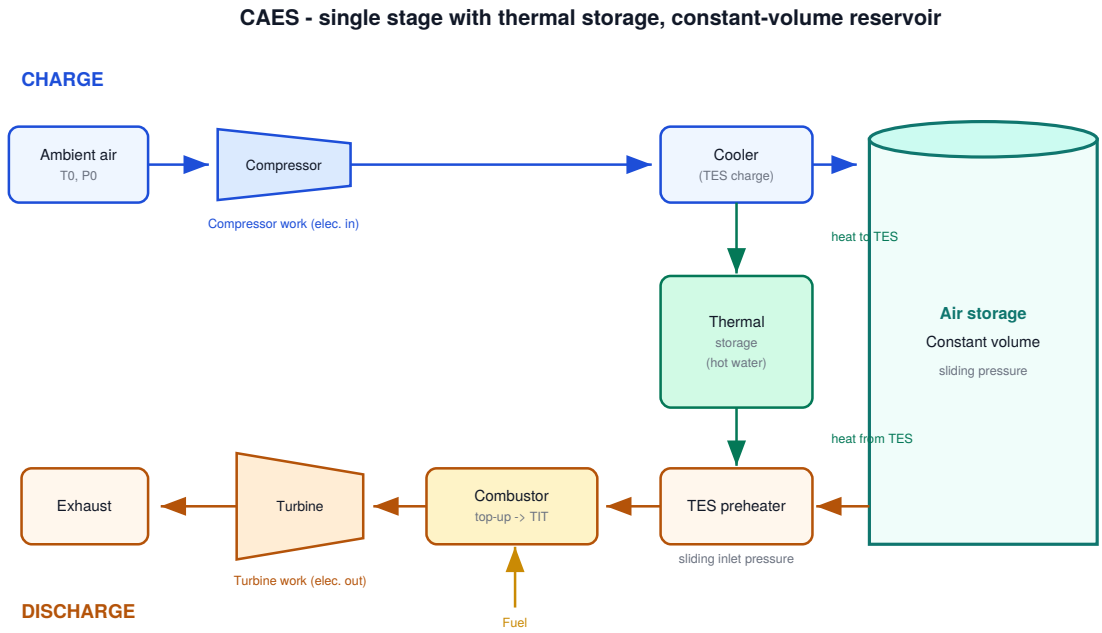
What is specific. One compression stage ($1 \to P_{\max}$) and one expansion stage. The cooling heat is rejected (not recovered). Constant-volume store: sliding pressure, turbine fed at the sliding store pressure. **Design.** Size the compressor, the cooler, the turbine, and the **storage volume** needed to deliver 4 MWh between $P_{\max}$ and $P_{\min}$. **Simulation.** Simulate the charge, then the discharge as a **transient** (store pressure falling from $P_{\max}$ to $P_{\min}$); evaluate η and the comparison criteria.

Variant 1b — single stage, no thermal storage, constant-pressure reservoir



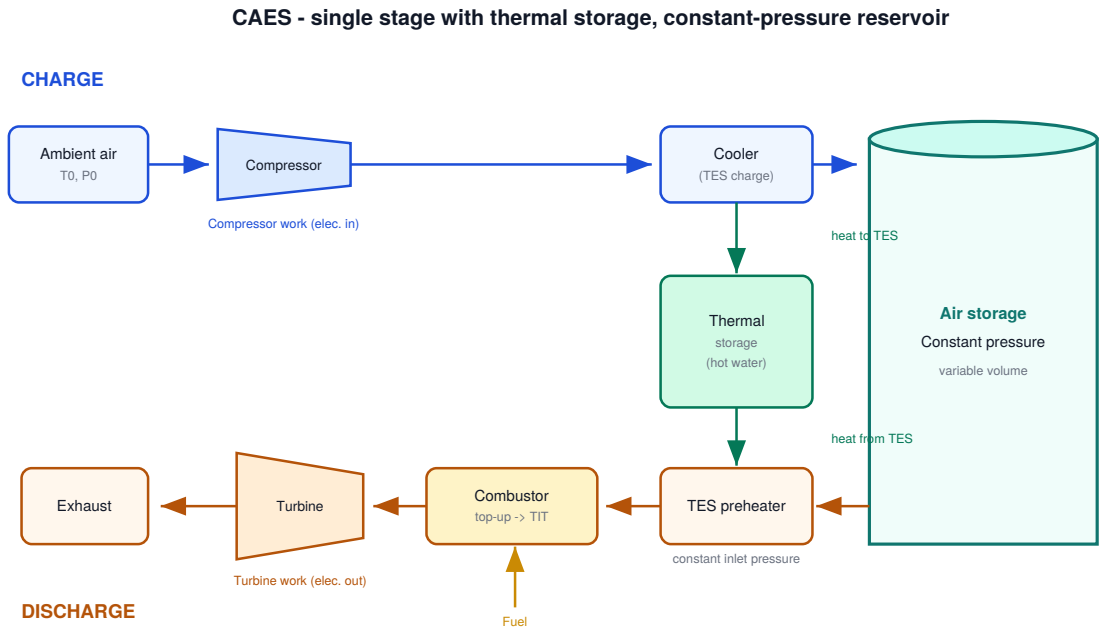
What is specific. Same architecture as 1a, but the store is held at constant pressure $P_{\max}$ (variable volume); the turbine sees a constant inlet pressure. The discharge is therefore quasi-steady. Compare the required volume and η against 1a.

Variant 2a — single stage with thermal storage, constant-volume reservoir



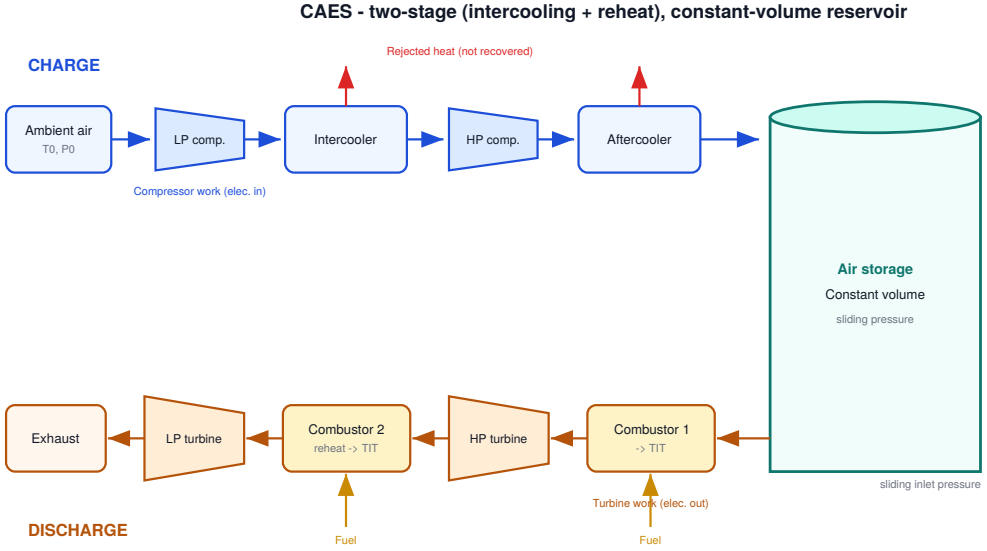
What is specific. The compression heat is **stored** (pressurised hot water, §3.2) during charge, and **reused** during discharge to preheat the air before combustion, so the combustion only tops up to the TIT. Model the thermal store (charging and discharging, with the 10 K return pinch). Note that only the reusable fraction is stored and a trim cooler rejects the rest (§3.3). Constant-volume store (sliding pressure). **Expected effect.** Lower fuel consumption than 1a for the same electricity delivered.

Variant 2b — single stage with thermal storage, constant-pressure reservoir



What is specific. As 2a, but with a constant-pressure air store (constant turbine inlet pressure). Isolates the reservoir-type effect when thermal storage is present.

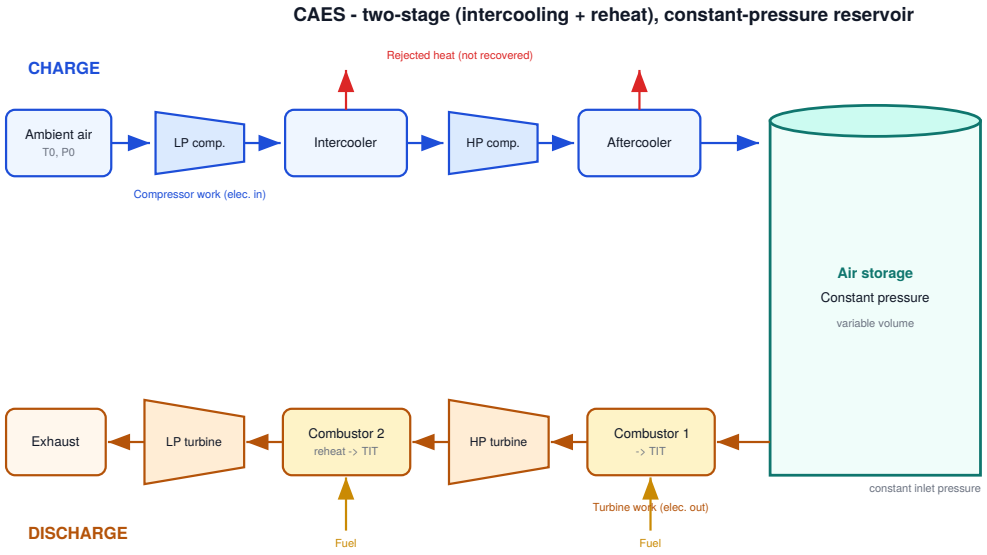
Variant 3a — two-stage (intercooling + reheat), no thermal storage, constant-volume reservoir



What is specific. Two compression stages with **intercooling** (equal pressure-ratio split) reduce the compression work; two expansion stages with **intermediate reheat** (combustion to the TIT before each turbine) increase the expansion work. The cooling heat is rejected (no thermal storage). Size the two coolers, the two combustors, the two turbines, and the storage volume. With the sliding

store pressure, the equal-split intermediate expansion pressure $P_{\text{int}} = \sqrt{p_{\text{store}}\, P_0}$ **slides in time**; state how it is controlled. **Expected effect.** Lower compressor work and higher turbine work than the single-stage cases.

Variant 3b — two-stage (intercooling + reheat), no thermal storage, constant-pressure reservoir



What is specific. As 3a, but with a constant-pressure air store (constant turbine inlet pressure). The intermediate pressure is fixed, $P_{\text{int}} = \sqrt{P_0\, P_{\max}} \approx 4.0$ bar.

5. Expected deliverables

Each group submits:

1. **A functional Python model** of its variant, built on the shared library:
   - **design mode** — components sized to the common specification (heat-exchanger areas, storage volume);
   - **simulation mode** — a full charge/discharge cycle, giving the performance metric η and the comparison criteria.
2. **An analysis of the results**: energy and (optionally) exergy balances, where the losses are, the split between electrical and fuel input, the required storage volume, and the maximum temperatures.
3. **At least one proposal to improve performance**, quantified with the model (e.g. effect of pressure level, of the number of stages, of the depth of discharge $P_{\min}$, of the thermal-storage pinch).
4. **Commented Python code**: comments and docstrings written as **general-purpose library documentation** (the library serves uses beyond this mini-project), stating assumptions and units.

Final cross-group comparison

Because the six variants share the specification and the fixed efficiencies, the groups pool their results and compare, on one chart, the **round-trip efficiency** η, the **required storage volume**, the **heat-exchanger sizes**, and the **electrical/fuel input split** — drawing conclusions on the value of thermal storage, of staging, and of the reservoir type.